

12. (a) Queen bees secrete a pheromone, compound **W**, which is used for a number of purposes, both inside and outside the hive.

Compound **W**

- is a straight chain aliphatic compound
- has a relative molecular mass of 184
- contains only carbon, hydrogen and oxygen
- contains 65.2% carbon and 26.1% oxygen by mass
- reacts with an alkaline solution of iodine to give a yellow solid
- produces effervescence when added to sodium hydrogencarbonate solution
- has a C=C double bond between carbons 2 and 3 and is an *E*-isomer
- produces an orange solid when reacted with 2,4-dinitrophenylhydrazine but does not give a silver mirror with Tollens reagent

Use all of this information to suggest a displayed formula for compound **W**. [6 QER]

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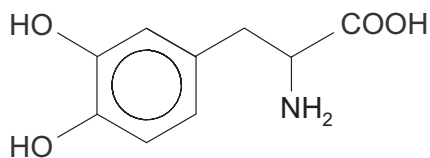
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- (b) Broad bean plants produce the α -amino acid L-dihydroxyphenylalanine (L-DOPA), which has important uses in the treatment of Parkinson's disease.



L-DOPA

- (i) Identify the chiral centre on the formula of L-DOPA by using an asterisk (*). [1]
- (ii) A solution containing equimolar proportions of L-DOPA and its enantiomer D-DOPA is described as a racemic mixture.

Explain why this mixture has no apparent effect on the plane of plane polarised light. [2]

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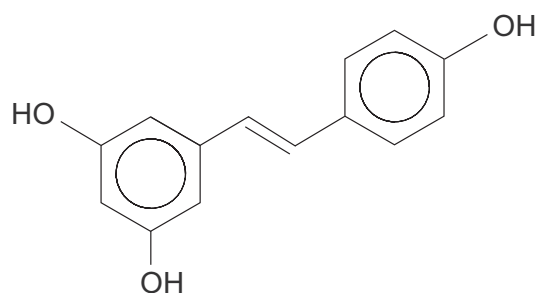
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- (iii) Write the displayed formula of the dipeptide formed from L-DOPA. [1]

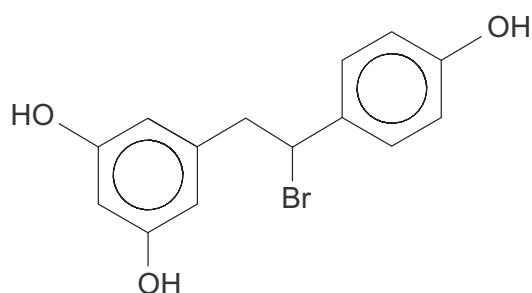


(c) Grape skins contain resveratrol.



resveratrol

- (i) The addition of hydrogen bromide to resveratrol results in the compound whose formula is shown below.



Explain why this addition does not involve the aromatic rings.

[1]

- (ii) Suggest a displayed formula for a compound produced when excess aqueous bromine is added to resveratrol. [1]

